

Pitching Evolution of College Baseball: A Machine Learning System for Modeling Stuff+, Location+, and Pitcher Development Using Division I TrackMan Data

Chris Jones

Department of Computer and Information Sciences

Fordham University

New York, USA

cjj12@fordham.edu

Abstract—College baseball programs now sit on top of an enormous amount of pitch-level data, but most of them do not have the tools or infrastructure to turn that data into something coaches can actually use. With TrackMan installed in nearly every Division I stadium, teams collect millions of pitches per season, yet the analytics systems that exist at the MLB level rarely translate cleanly to the college game. This project builds a complete machine learning system tailored specifically to Division I baseball, using more than ten million pitch-by-pitch observations collected between 2022 and the present. The dataset, consisting of more than 10,000 individual TrackMan CSV files, was obtained through my work with the Fordham baseball coaching staff and transferred securely from their FileZilla server using a custom Python ingestion script. TrackMan, a Doppler radar system now integrated into nearly every Division I stadium (and increasingly at lower levels), provides high-resolution measurements of pitch movement, release metrics, and outcomes.

Two machine learning models were developed: a Stuff+ model that predicts the probability of generating a called or swinging strike (CSW) using ball-flight characteristics, and a Location+ model that predicts the expected run value of a pitch based on its location and count context. Both models use LightGBM and scale predictions relative to Division I averages to produce interpretable “plus” metrics centered at 100. Alongside the models, I built a complete data pipeline, feature engineering framework, and a Streamlit Cloud application that provides interactive visualizations, single-game summaries, and full pitcher profiles with season stats and underlying metrics. The results show how machine learning can enhance player development, surface under-utilized pitches, and support evidence-based coaching decisions in a modern collegiate baseball environment.

Index Terms—Baseball analytics, machine learning, TrackMan, Stuff+, Location+, LightGBM, sports analytics, pitcher development

I. INTRODUCTION

Over the past decade, baseball has shifted from being driven mostly by radar guns and scouting notes to being driven by high-resolution tracking systems and large-scale data. TrackMan, a Doppler radar platform that measures pitch velocity, spin, movement, release metrics, and ball flight, has become the standard across Major League Baseball and is now installed in nearly every Division I NCAA stadium. As a

result, college programs generate millions of data points each season, enabling a level of quantitative evaluation of pitcher performance that simply did not exist a decade ago.

Major League organizations have aggressively adopted advanced metrics such as Stuff+, Location+, and Pitching+, using them to evaluate pitchers beyond traditional statistics like ERA or WHIP. In contrast, many collegiate programs still rely heavily on those traditional stats and subjective evaluation, even though they now have access to the same underlying tracking technology. The gap between the amount of data available and the ability to turn it into actionable insight is what motivates this work.

This project leverages more than 10 million TrackMan-tracked pitches from Division I baseball (2022–present). Working closely with the Fordham coaching staff directly shaped the design of the system. The coaches repeatedly pointed out that MLB-level models, when applied directly to college data, often produce misleading “plus” scores that do not reflect the true distribution of collegiate pitch quality. Average Division I pitches can look far below average simply because the reference population is different.

To address this, I designed and implemented a complete machine learning system that computes Stuff+ and Location+ metrics calibrated specifically to the Division I environment. The system includes a robust ingestion pipeline, extensive feature engineering, model training and evaluation, and a deployed Streamlit Cloud application that exposes interactive dashboards to coaches and players. The overarching goal is to translate raw TrackMan data into interpretable, actionable metrics that can drive real changes in pitch usage, development plans, and game strategy.

II. RELATED WORK

The concept of Stuff+ was popularized in the public baseball analytics community by Driveline Baseball and FanGraphs. Driveline’s 2024 revisiting of Stuff+ emphasizes that pitch quality can be modeled using velocity, movement, and spin-based characteristics, independent of results, by focusing on

the physical properties of the pitch rather than outcomes that are heavily influenced by context and randomness [1]. FanGraphs provides a detailed primer on Stuff+, Location+, and Pitching+, explaining how these models quantify raw stuff, command, and overall effectiveness in a unified framework [2].

Collegiate baseball research has started to adapt these ideas to non-professional environments. The CornBelters’ analysis of Stuff+ in college settings shows how Stuff+ can identify pitchers with elite movement profiles even when traditional results lag behind [3]. Rockland Peak Performance (RPP) further explains how Stuff+ can guide pitch design, helping pitchers understand which pitch shapes are most effective and how to optimize their arsenal through targeted adjustments to grip, release, and intent [4].

However, most publicly documented implementations of Stuff+ and Location+ are calibrated to MLB-level data distributions. Applying these models directly to college environments can lead to systematic mis-scaling, where average Division I pitches are scored as significantly below average simply because the underlying reference population is different. This project extends prior work by (1) building models on a large-scale Division I dataset, (2) calibrating “plus” metrics relative to collegiate distributions, and (3) integrating the results into a real-time analytics platform tailored specifically to Fordham Baseball.

III. DATA DESCRIPTION

A. TrackMan Technology

TrackMan uses Doppler radar to track the baseball from release to home plate, capturing a rich set of ball-flight and release metrics. For each pitch, the system records:

- release height and side,
- velocity,
- spin rate and spin axis,
- induced vertical break (IVB),
- horizontal break (HB),
- plate location (both horizontal and vertical),
- pitch result (ball, strike, in-play outcome, etc.).

TrackMan has been integrated into nearly every Division I NCAA baseball stadium, making it the primary source of pitch-level data at the collegiate level. The consistency of the technology across venues enables cross-team and cross-conference comparisons that were previously impossible.

B. Data Acquisition

The Fordham coaching staff provided private access to their TrackMan FileZilla server, which stores pitch-by-pitch CSV exports for every game and practice session. I developed a custom Python script to:

- connect securely to the FTP server,
- recursively scan all directories for CSV files,
- download every CSV file to a local machine,
- validate file integrity (e.g., file size, basic schema checks),
- organize files into a structured local dataset by season, team, and game.

More than 10,000 CSV files were transferred, requiring several hours of automated downloading. The final dataset contains over 10 million individual pitch records spanning multiple seasons and opponents.

C. Data Privacy and Access Restrictions

Because the TrackMan dataset contains proprietary information from Fordham and other Division I programs, the raw CSV files cannot be shared publicly. Only aggregated results, derived features, and model outputs are included in the public GitHub repository associated with this project. This constraint influences the design of the system: all sensitive processing is performed locally or within controlled environments, while the deployed Streamlit application exposes only summary-level analytics.

D. Absence of Biomechanical Data

TrackMan does not capture biomechanical measurements such as arm slot variability, torso tilt, hip-shoulder separation, stride length, or joint kinetics. As a result, the models in this project evaluate pitch quality strictly from ball-flight characteristics and command. While this is sufficient for computing Stuff+ and Location+, it limits the ability to directly connect mechanical changes to changes in pitch quality. Future integration with motion capture or wearable sensor data could help bridge this gap.

E. Why Stuff+ and Location+ Matter

Stuff+ and Location+ provide a quantitative framework for:

- identifying pitches with elite movement or command,
- detecting under-utilized pitches that may deserve higher usage,
- evaluating pitch quality independent of noisy outcomes,
- supporting pitch design and individualized development plans,
- benchmarking pitchers against Division I peers rather than MLB populations.

By centering these metrics at 100 for the Division I population, coaches can quickly interpret whether a given pitch or pitcher is below average, average, above average, or truly elite within the collegiate context.

IV. METHODOLOGY

A. Feature Engineering

The system computes more than 20 derived features for each pitch, including:

- CSW% (called strikes + swinging strikes divided by total pitches),
- Whiff% (swinging strikes divided by swings),
- Strike% (all strikes divided by total pitches),
- Zone% (pitches in the strike zone divided by total pitches),
- release drift (difference between release location and plate location),
- movement clusters (clustering IVB and HB to identify pitch shape families),

- batter-handedness splits (left-handed vs. right-handed batters),
- count leverage indicators (e.g., hitter-friendly vs. pitcher-friendly counts).

These features are engineered both at the pitch level and aggregated at the pitch-type and pitcher levels to support downstream modeling and visualization.

B. Stuff+ Model (CSW-Based)

1) *Target Variable*: Stuff+ predicts the probability that a pitch results in a called or swinging strike (CSW). We define the binary target as:

$$\text{CSW} = 1\{\text{StrikeCalled or StrikeSwinging}\}.$$

This formulation focuses the model on the ability of a pitch to generate strikes through either deception (whiffs) or command (called strikes), independent of batted-ball outcomes.

2) *Features*: The Stuff+ model uses pure ball-flight physics and release metrics:

- velocity (Velo),
- induced vertical break (IVB),
- horizontal break (HB),
- spin rate (Spin),
- release height (RelH),
- release side (RelS),
- extension (Ext),
- vertical approach angle (VAA),
- horizontal approach angle (HAA).

By excluding contextual variables such as count and batter quality, the model isolates the intrinsic quality of the pitch itself.

3) *Model*: A LightGBM classifier is trained on millions of pitches. LightGBM is a gradient boosting framework that is well-suited for large tabular datasets and can handle non-linear interactions between features efficiently [5]. Hyperparameters such as learning rate, maximum depth, number of leaves, and regularization terms are tuned using cross-validation on a held-out subset of the data.

4) *Scaling*: Raw predicted probabilities are transformed into a “plus” metric using:

$$\text{Stuff+} = 100 + 50 \cdot \frac{p - \mu}{\sigma},$$

where p is the predicted CSW probability for a given pitch, μ is the league mean predicted CSW across all Division I pitches in the training set, and σ is the corresponding standard deviation. This scaling ensures that 100 represents the Division I average, 120 represents roughly two standard deviations above average, and 80 represents roughly two standard deviations below average.

C. Location+ Model (Run-Value-Based)

1) *Target Variable*: Location+ predicts the expected run value of a pitch based on its location and count context. We define:

$$RV = f(\text{PlayResult}),$$

where f maps each play result (e.g., home run, single, strikeout, walk) to a run value using a standard run expectancy framework (e.g., HR = +1.40, strikeout = -0.29). This target captures the impact of pitch location on run prevention.

2) *Features*: The Location+ model uses:

- plate location (plate_x, plate_z),
- zone (discrete strike zone region),
- balls and strikes (count),
- pitch type (categorical encoded),
- handedness matchup (optional extension).

By focusing on location and count, the model isolates the command component of pitching, separate from raw stuff.

3) *Model*: A LightGBM regressor predicts the expected run value of each pitch. As with the Stuff+ model, hyperparameters are tuned via cross-validation. Because run value is noisy at the pitch level, the model is evaluated primarily on its ability to outperform simple baselines and capture meaningful patterns rather than achieving high R^2 values.

4) *Scaling*: Location+ is scaled analogously to Stuff+:

$$\text{Location+} = 100 + 50 \cdot \frac{RV - \mu}{\sigma},$$

where RV is the predicted run value for a given pitch, μ is the league mean predicted run value, and σ is the standard deviation. Higher Location+ scores correspond to better (more negative) expected run values, indicating superior command.

D. Evaluation Metrics

To assess the performance of both the Stuff+ (CSW classifier) and Location+ (run-value regressor) models, we compute standard evaluation metrics appropriate for each prediction task. Metrics are calculated on a held-out test set that is not used during training or hyperparameter tuning.

1) *Stuff+ Model*: The Stuff+ model predicts the probability that a pitch results in a called or swinging strike (CSW). Because CSW events are relatively rare in Division I datasets, the class distribution is highly imbalanced, making ROC-AUC and log loss more informative than raw accuracy.

Results:

- Accuracy: 0.7200
- Precision: 0.6535
- Recall: 0.0029
- F1 Score: 0.0058
- ROC-AUC: 0.6996
- Log Loss: 0.5406

The low recall reflects the fact that the positive class (CSW = 1) is rare and that the classifier is conservative in assigning high probabilities. However, Stuff+ is used as a continuous probability-based metric rather than a hard classifier, so recall is less critical than calibration and ranking performance. The ROC-AUC of approximately 0.70 indicates that the model can meaningfully distinguish between pitches that are more or less likely to generate CSW, consistent with published Stuff+ methodologies.

2) *Location+ Model*: The Location+ model predicts the expected run value of a pitch based on its location, count, and pitch type. Run value is inherently noisy and influenced by batter quality, defense, and randomness, so modest R^2 values are typical even in professional baseball models.

Results:

- RMSE: 0.1630
- MAE: 0.0672
- R^2 : 0.0690
- Baseline RMSE (predict league mean): 0.1689
- Baseline MAE: 0.0717

The model outperforms the baseline on both RMSE and MAE, demonstrating that it captures meaningful relationships between pitch location, count leverage, and run value. The R^2 value (≈ 0.07) is consistent with expectations for pitch-level run-value prediction, where noise dominates individual outcomes. These results confirm that the Location+ model provides a statistically valid and practically useful measure of command quality.

V. RESULTS

A. Total Stuff+ and Location+ Leaderboards

Total Stuff+ and Location+ leaderboards were generated to rank pitchers across the dataset. These leaderboards provide a high-level view of overall pitch quality (Stuff+) and command (Location+) at the Division I level and serve as an entry point for deeper analysis.

The Stuff+ leaderboard (Fig. 1) highlights pitchers whose arsenals generate above-average CSW rates purely from ball-flight characteristics. The Location+ leaderboard (Fig. 2) highlights pitchers who consistently locate pitches in zones that minimize expected run value, even if their raw stuff is less dominant.

B. Pitch-Type Stuff+ and Location+ Leaderboards

Pitch-type-specific leaderboards were computed for changeups (CH), curveballs (CU), fastballs (FB), cutters (FC), sinkers (SI), and sliders (SL). These leaderboards highlight which individual pitch types exhibit elite movement or command, allowing coaches to identify strengths and weaknesses within a pitcher's arsenal.

Figs. 3 and 4 reveal that some pitchers possess individual pitches (e.g., a slider or changeup) that rank among the best in Division I, even if their overall Stuff+ or Location+ is closer to average. This motivates more nuanced usage recommendations that focus on optimizing pitch mix rather than treating all pitches as interchangeable.

C. Interpreting Stuff+ and Location+ Scores

Stuff+ and Location+ are scaled so that 100 corresponds to the Division I average.

Stuff+ scale:

- < 85: below average
- 85–100: slightly below average
- 100–115: above average
- 120+: elite

Fordham Baseball — Total Stuff+

Berg, Aric	116.8
Dowd, Aidan	114.3
Hanawalt, Chase	110.8
Kapica, Andrew	110.4
Cawley, Declan	110.4
Smith, Koen	110.3
Redick, James	103.1
Stewart, Robbie	102.9
Reilly-Bell, Diego	100.8
Egan, Ryan	99.9
Murray, Alec	97.8
Elson, Beau	97.5
Osterhus, Eric	95.4
Rodarte, Jordan	94.8
Chavez, Caden	94.4
Travaglia, Giacomo	92.7
O'Brien, Aidan	91.9
Scarlata, Anthony	87.9
Morello, Mason	82.4
Vieira, William	78.5
Clark, Sam	69.0

Fig. 1. Total Stuff+ leaderboard for Division I pitchers. Higher values indicate better raw pitch quality relative to the Division I average.

Location+ scale:

- < 90: poor command
- 90–100: average
- 100–110: above average
- 115+: elite

These thresholds provide coaches with simple interpretive bands. For example, a pitch with Stuff+ = 122 and Location+ = 108 would be considered both elite in raw quality and above average in command, making it a strong candidate for increased usage in high-leverage situations.

D. Example: Identifying Under-Utilized Pitches (Alec Murray)

Fig. 5 shows a full pitching profile visualization for Alec Murray, a Fordham pitcher whose arsenal includes a fastball and a cutter among other pitches.

Key insights from the profile include:

- Fastball usage: 45.8% (Stuff+ = 78.48, below average).

Fordham Baseball — Total Location+		
Elson, Beau	105.9	
Cawley, Declan	105.8	
Stewart, Robbie	102.5	
Berg, Aric	102.3	
Dowd, Aidan	101.6	
Murray, Alec	101.5	
Kapica, Andrew	101.2	
Rodarte, Jordan	100.8	
Hanawalt, Chase	99.7	
O'Brien, Aidan	98.9	
Egan, Ryan	98.3	
Vieira, William	96.1	
Chavez, Caden	96.1	
Scarlata, Anthony	93.9	
Smith, Koen	93.7	
Redick, James	92.0	
Travaglia, Giacomo	91.1	
Morello, Mason	86.8	
Reilly-Bell, Diego	83.4	
Osterhus, Eric	80.0	
Clark, Sam	78.1	

Fig. 2. Total Location+ leaderboard for Division I pitchers. Higher values indicate better command and more favorable expected run values.

- Cutter usage: 18.9% (Stuff+ = 114.52, elite/above average).
- The cutter also exhibits competitive Location+ relative to the fastball.

The model suggests increasing cutter usage to improve performance, as the cutter exhibits significantly better Stuff+ than the fastball despite lower usage. This is a concrete example of how the system can identify under-utilized pitches and support evidence-based adjustments to pitch mix.

VI. SYSTEM DESIGN

A. Architecture

The overall system architecture consists of the following stages:

- **FileZilla ingestion:** automated retrieval of TrackMan CSV files from the FTP server.
- **Feature engineering:** computation of derived metrics and aggregation at multiple levels.
- **Model inference:** application of trained Stuff+ and Location+ models to new data.
- **Streamlit multi-page interface:** interactive dashboards for coaches and players.

CH - Top 10 Stuff+			CU - Top 10 Stuff+			FB - Top 10 Stuff+		
Dowd, Aidan	St+: 109.0 LHH: 103.7 RHH: 114.0		Hanawalt, Chase	St+: 129.3 LHH: 142.8 RHH: 122.6		Cawley, Declan	St+: 116.8 LHH: 105.6 RHH: 115.5	
Stewart, Robbie	St+: 98.0 LHH: 101.8 RHH: 93.6		Berg, Aric	St+: 128.0 LHH: 108.7 RHH: 125.4		Egan, Ryan	St+: 104.9 LHH: 112.1 RHH: 102.3	
Hanawalt, Chase	St+: 95.2 LHH: 90.9 RHH: 131.6		Kapica, Andrew	St+: 121.1 LHH: 129.2 RHH: 116.9		Berg, Aric	St+: 103.4 LHH: 120.4 RHH: 99.5	
Egan, Ryan	St+: 94.9 LHH: 97.7 RHH: 93.1		Elson, Beau	St+: 117.0 LHH: 136.0 RHH: 116.0		Chavez, Caden	St+: 98.2 LHH: 108.0 RHH: 94.1	
Cawley, Declan	St+: 92.2 LHH: 89.8 RHH: 90.7		Chavez, Caden	St+: 103.3 LHH: 142.0 RHH: 98.8		Hanawalt, Chase	St+: 97.1 LHH: 90.7 RHH: 101.4	
Kapica, Andrew	St+: 92.2 LHH: 89.8 RHH: 96.9		Dowd, Aidan	St+: 100.9 LHH: 98.3 RHH: 109.6		Kapica, Andrew	St+: 96.6 LHH: 96.2 RHH: 96.8	
Rodarte, Jordan	St+: 89.6 LHH: 85.9 RHH: 95.5		Egan, Ryan	St+: 88.8 LHH: 111.3 RHH: 83.2		O'Brien, Aidan	St+: 94.4 LHH: 92.7 RHH: 94.7	
Scarlata, Anthony	St+: 88.5 LHH: 89.7 RHH: 86.2		Vieira, William	St+: 78.0 LHH: 76.9 RHH: 78.2		Osterhus, Eric	St+: 94.3 LHH: 152.7 RHH: 92.0	
Chavez, Caden	St+: 86.5 LHH: 74.8 RHH: 87.2					Rodarte, Jordan	St+: 94.0 LHH: 93.4 RHH: 94.6	
Elson, Beau	St+: 82.8 LHH: nan RHH: 82.8					Elson, Beau	St+: 92.3 LHH: 83.3 RHH: 95.7	
FC - Top 10 Stuff+			SI - Top 10 Stuff+			SL - Top 10 Stuff+		
Dowd, Aidan	St+: 129.9 LHH: 127.7 RHH: 130.3		Stewart, Robbie	St+: 100.3 LHH: 89.3 RHH: 106.2		Reilly-Bell, Diego	St+: 140.5 LHH: nan RHH: 140.5	
Hanawalt, Chase	St+: 121.1 LHH: 102.8 RHH: 124.1		Travaglia, Giacomo	St+: 90.0 LHH: 81.2 RHH: 109.3		Berg, Aric	St+: 134.4 LHH: 137.4 RHH: 133.8	
Stewart, Robbie	St+: 119.7 LHH: 116.9 RHH: 120.3		Morello, Mason	St+: 88.9 LHH: 88.6 RHH: 89.0		Dowd, Aidan	St+: 129.5 LHH: 127.8 RHH: 126.8	
Murray, Alec	St+: 116.8 LHH: 124.7 RHH: 113.6		Reilly-Bell, Diego	St+: 86.1 LHH: nan RHH: 86.1		Kapica, Andrew	St+: 127.5 LHH: 127.4 RHH: 127.4	
Elson, Beau	St+: 92.4 LHH: 81.4 RHH: 97.4		Kapica, Andrew	St+: 77.6 LHH: 42.1 RHH: 94.2		Smith, Koen	St+: 127.1 LHH: 126.0 RHH: 127.6	
Scarlata, Anthony	St+: 85.8 LHH: 72.7 RHH: 95.7					Hanawalt, Chase	St+: 126.4 LHH: 147.5 RHH: 118.6	
						Elson, Beau	St+: 116.9 LHH: 117.3 RHH: 116.9	
						Redick, James	St+: 115.1 LHH: 135.2 RHH: 109.2	
						O'Brien, Aidan	St+: 112.9 LHH: nan RHH: 112.9	
						Murray, Alec	St+: 112.8 LHH: 125.7 RHH: 108.1	

Fig. 3. Top 10 Stuff+ scores by pitch type (CH, CU, FB, FC, SI, SL). Each bar represents a pitcher-pitch-type combination.

This modular design allows each component to be updated independently. For example, new features can be added to the engineering stage without changing the ingestion logic, and models can be retrained and redeployed without modifying the Streamlit front-end.

B. Deployment

The system is deployed on Streamlit Cloud, which hosts the application and handles user access. Key deployment considerations include:

- cached model loading to minimize latency,
- efficient data loading and filtering for large datasets,
- PDF report generation for post-game summaries and scouting packets,
- environment configuration to ensure compatibility with LightGBM and other dependencies.

By deploying to Streamlit Cloud, the coaching staff can access the application from any device with a web browser, without needing to manage local installations.

C. Automated Post-Game Updating Workflow

The automated workflow for updating analytics after each game is:

- coaches export TrackMan CSV files from the stadium system,

CH - Top 10 Loc+		CU - Top 10 Loc+		FB - Top 10 Loc+	
Stewart, Robbie	Loc+: 111.7 LHH: 121.9 RHH: 100.1	Dowd, Aidan	Loc+: 114.4 LHH: 108.5 RHH: 134.0	Kapica, Andrew	Loc+: 111.3 LHH: 112.0 RHH: 111.0
Cawley, Declan	Loc+: 111.1 LHH: 109.4 RHH: 112.2	Vieira, William	Loc+: 100.0 LHH: 111.6 RHH: 97.6	Elson, Beau	Loc+: 108.5 LHH: 104.5 RHH: 109.5
Chavez, Caden	Loc+: 108.9 LHH: 67.4 RHH: 111.3	Berg, Aric	Loc+: 94.3 LHH: 91.3 RHH: 85.4	Berg, Aric	Loc+: 106.6 LHH: 125.2 RHH: 102.4
Dowd, Aidan	Loc+: 106.0 LHH: 111.1 RHH: 101.3	Elson, Beau	Loc+: 90.6 LHH: 73.7 RHH: 91.4	Stewart, Robbie	Loc+: 106.2 LHH: 100.3 RHH: 109.1
Scarlata, Anthony	Loc+: 95.3 LHH: 84.7 RHH: 121.2	Hanawalt, Chase	Loc+: 90.5 LHH: 86.4 RHH: 92.5	Cawley, Declan	Loc+: 106.0 LHH: 102.0 RHH: 107.0
Rodarte, Jordan	Loc+: 95.5 LHH: 90.6 RHH: 103.9	Kapica, Andrew	Loc+: 89.4 LHH: 83.7 RHH: 87.2	Dowd, Aidan	Loc+: 105.2 LHH: 84.7 RHH: 114.4
Egan, Ryan	Loc+: 94.1 LHH: 92.6 RHH: 94.9	Chavez, Caden	Loc+: 84.8 LHH: 85.6 RHH: 87.5	Rodarte, Jordan	Loc+: 104.7 LHH: 102.7 RHH: 106.4
Kapica, Andrew	Loc+: 93.0 LHH: 86.7 RHH: 102.7	Egan, Ryan	Loc+: 70.3 LHH: 83.2 RHH: 67.1	Egan, Ryan	Loc+: 104.2 LHH: 108.2 RHH: 102.7
Elson, Beau	Loc+: 92.5 LHH: nan RHH: 92.5			Hanawalt, Chase	Loc+: 103.9 LHH: 96.6 RHH: 106.8
O'Brien, Aidan	Loc+: 90.8 LHH: 92.8 RHH: 89.3			Murray, Alec	Loc+: 103.6 LHH: 101.4 RHH: 105.1
FC - Top 10 Loc+		SI - Top 10 Loc+		SL - Top 10 Loc+	
Elson, Beau	Loc+: 122.0 RHH: 112.6 LHH: 120.7	Travaglia, Giacomo	Loc+: 107.6 LHH: 89.1 RHH: 148.2	Murray, Alec	Loc+: 100.9 LHH: 109.4 RHH: 97.8
Stewart, Robbie	Loc+: 113.2 LHH: 97.3 RHH: 116.4	Stewart, Robbie	Loc+: 94.0 LHH: 84.0 RHH: 92.4	Berg, Aric	Loc+: 100.6 LHH: 91.3 RHH: 104.1
Dowd, Aidan	Loc+: 100.5 LHH: 104.7 RHH: 99.6	Kapica, Andrew	Loc+: 93.0 LHH: 71.7 RHH: 102.9	Hanawalt, Chase	Loc+: 100.1 LHH: 115.5 RHH: 95.1
Hanawalt, Chase	Loc+: 99.9 LHH: 100.7 RHH: 99.8	Morello, Mason	Loc+: 91.5 LHH: 86.9 RHH: 92.6	Kapica, Andrew	Loc+: 98.0 LHH: 105.2 RHH: 94.1
Murray, Alec	Loc+: 98.2 LHH: 108.0 RHH: 94.7	Reilly-Bell, Diego	Loc+: 85.3 LHH: nan RHH: 85.3	O'Brien, Aidan	Loc+: 97.7 LHH: nan RHH: 97.7
Scarlata, Anthony	Loc+: 91.9 LHH: 86.5 RHH: 95.9			Stewart, Robbie	Loc+: 96.8 LHH: 122.2 RHH: 91.2
				Cawley, Declan	Loc+: 95.1 LHH: 50.9 RHH: 95.7
				Dowd, Aidan	Loc+: 95.0 LHH: 69.7 RHH: 96.3
				Elson, Beau	Loc+: 94.9 LHH: 97.5 RHH: 93.9
				Smith, Koen	Loc+: 93.2 LHH: 84.3 RHH: 96.5

Fig. 4. Top 10 Location+ scores by pitch type (CH, CU, FB, FC, SI, SL). Higher scores indicate superior command for that pitch type.

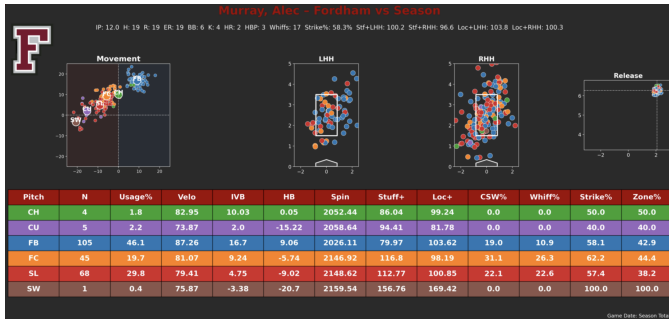


Fig. 5. Alec Murray pitching profile visualization, including Stuff+ and Location+ by pitch type and usage rates.

- CSV files are placed in a designated GitHub /data folder,
- a scheduled process or manual trigger updates the processed dataset,
- Streamlit automatically reflects the new data the next time the app is loaded.

This workflow minimizes friction for the coaching staff and ensures that analytics are available shortly after each game, enabling timely feedback and adjustments.

VII. DISCUSSION

A. Data Privacy Constraints

Raw TrackMan CSVs cannot be shared publicly due to NCAA and program-level restrictions. This limits external reproducibility and the ability of third parties to validate the models directly. However, the methodology, code, and aggregated outputs can still be shared, providing transparency into the modeling approach while respecting privacy constraints.

B. Limitations

Key limitations of the current system include:

- **No biomechanical data:** the models cannot directly account for mechanical changes or inefficiencies.
- **TrackMan noise:** measurement error in velocity, spin, and movement can introduce noise into the features.
- **No pitch intent labels:** the system does not know whether a pitch was executed as intended (e.g., missed location vs. deliberate waste pitch).
- **No run-value modeling for Stuff+:** Stuff+ is based on CSW rather than run value, which may not capture all aspects of pitch effectiveness.
- **No batter quality adjustment:** Location+ does not currently adjust for the quality of opposing hitters, which can influence run value.

These limitations suggest caution when interpreting individual pitch or game-level metrics, especially in small samples.

C. Future Work

Future directions for extending this work include:

- **Command+ modeling:** integrating both location and intent to better capture true command.
- **Pitch tunneling analysis:** quantifying how well different pitches tunnel out of the hand before diverging.
- **Full run-value modeling:** building models that directly predict run value for Stuff+ as well as Location+.
- **Biomechanics integration:** combining TrackMan data with motion capture or wearable sensors.
- **Automated scouting reports:** generating natural-language summaries of pitcher strengths, weaknesses, and recommended usage patterns.

These extensions would further enhance the system's value as a comprehensive player development and scouting tool.

VIII. CONCLUSION

This project delivers a complete machine learning system for evaluating Fordham's pitching performance using advanced analytics and large-scale TrackMan data. By computing Division I-calibrated Stuff+ and Location+ metrics and exposing them through an interactive Streamlit application, the system bridges the gap between raw tracking data and actionable coaching insight.

The combination of robust data ingestion, thoughtful feature engineering, well-calibrated models, and intuitive visualizations provides a powerful tool for player development. The case study of Alec Murray demonstrates how the system can

identify under-utilized pitches and suggest concrete adjustments to pitch mix. More broadly, this work illustrates how modern machine learning techniques can be applied to collegiate sports environments in a way that respects data privacy while still delivering meaningful competitive advantages.

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